■ MEADOWLANDS — ARCHIVED FEATURES INVENTORY

Comprehensive Audit of Dead Code & Subsystems Moved to archive/dead-code-backup
Generated: October 25, 2026 | **Repository:** abhi543/meadowlands | **Total Archived Code:** 22 Files • 9,409 Lines Removed

Executive Summary: In accordance with repository optimization goals, inactive, unreferenced, and deprecated subsystems were pruned from the active production branch (main) and consolidated into the dedicated archive branch **archive/dead-code-backup** (commit dbe5b25). This eliminated **9,409 lines** of dead code across 22 files, reducing bundle weight, improving build latency, and eliminating mechanics that created unnecessary friction (such as the 3-color soil battery penalties).

■ Summary Matrix of Archived Features

Subsystem / Feature	Category	Files Affected	Lines	Archival Rationale
Wilderness 3D Exploration	Mini-Game	3 files (WildExpedition3D.tsx, etc.)	**2,739**	Unreferenced 3D canvas; heavy Three.js bundle cost.
Estate Dispatch Center	Logistics	1 file (EstateDispatchCenterModal.tsx)	**1,820**	Legacy delivery mission UI superseded by unified dispatch.
Sluice Gate Control Console	Hydraulics	1 file (SluiceGateControlModal.tsx)	**1,052**	Redundant manual valve console; hydraulics automated.
3-Color Soil Battery & Tired Soil	Agronomy	5 files (soilEnergyEngine.ts, Grid, etc.)	**~780**	Depleted soil penalties, mini-bars & burnout removed.
Estate Menagerie Showcase	Sanctuary	1 file (EstateMenagerieModal.tsx)	**763**	Merged into unified Prehistoric Sanctuary Panel.
Crisis Defense & Alert Suite	Events	4 files (CrisisDefenseModal.tsx, etc.)	**727**	Unused standalone crisis modals & emergency clock widgets.
High-Stakes Contract Deck	Economy	1 file (HighStakesContractDeck.tsx)	**578**	Unreferenced futures contracts card interface.
Developer Test Suites	Dev / Test	3 files (server.ts, overflow.ts)	**568**	Stand-alone websocket and overflow scripts outside app.
Hydrology Gauges & Offline Modal	Hydraulics	2 files (EstateHydrologyGauge.tsx, etc.)	**448**	Duplicate analog dials & offline catchup prompt.
Legacy Placeholder Stubs	Routing	2 files (AnimalPanel.tsx, WildTab.tsx)	**16**	Obsolete empty stub components.
TOTALS	**All Systems**	**22 Files**	**9,409**	**Safely preserved on archive branch**

■ Detailed Architectural Breakdown of Archived Features

1. Wilderness 3D Exploration & Safari Expedition System [*2,739 lines*]

Overview: A complete interactive 3D safari and wildlife capture subsystem built with Three.js/Canvas. Players were able to deploy expeditions into procedural biomes (Taiga Forest, Volcanic Crag, Savannah) with dynamic fog-of-war, roving megafauna track scouting, tranquilizer dart ballistics, cage deployment traps, and captive specimen containment transport.
Reason for Removal: Three.js rendering added substantial client-side bundle overhead. The gameplay was siloed away from core farmstead loop mechanics and was replaced by the streamlined Prehistoric Sanctuary expedition system.

Archived Files:
• src/components/game/WildExpedition3D.tsx (1,573 lines)
• src/components/game/WildernessPanel.tsx (1,153 lines)
• src/components/game/WildTab.tsx (13 lines)

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2. Estate Task & Logistics Dispatch Center [*1,820 lines*]

Overview: A large-scale logistics orchestration modal managing estate courier routes, bulk freight contracts, cross-sector wagon logistics, export tariffs, and worker priority dispatching.
Reason for Removal: The modal was unreferenced in the main UI navigation. Task management was unified into the modern Master Worker (v2.0) automation system and the high-urgency Event Engine.

Archived Files:
• src/components/game/EstateDispatchCenterModal.tsx (1,820 lines)

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3. Sluice Gate Control Console & Canal Hydro-Valves [*1,052 lines*]

Overview: An elaborate multi-canal manual hydraulics console with pressure wheels, weir gate actuators, canal spillway emergency overflow levers, and individual gate percentage regulators across 4 estate canals.
Reason for Removal: Hydraulic simulation was refactored into the automated Gravity Tower and hydraulic fail-safe engine, rendering the manual 1,000-line mechanical valve modal redundant.

Archived Files:
• src/components/game/SluiceGateControlModal.tsx (1,052 lines)

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4. 3-Color Soil Battery & Tired Soil Mechanics [*~780 lines pruned*]

Overview: The color-coded agronomy battery system featuring Green (Leaf/Stem), Yellow (Root/Bulb), and Red (Fruit/Flower) nutrient energy meters. Crops planted without matching battery channels were flagged as 'Tired Soil', suffering 2x growth duration and being clamped at 1x harvest yield. Heatwaves caused battery burnout, while torrential rain leached nutrients.
Reason for Removal: Per user directive, the battery meters created excessive visual clutter (mini-bars over every plot) and punitive penalties that clashed with modern soil gas and dynamic moisture mechanics.

Archived Files:
• src/game/soilEnergyEngine.ts (178 lines)
• src/components/game/FarmGrid.tsx (310 lines modified/pruned)
• src/game/store.ts (190 lines modified/pruned)
• src/components/game/GuidePanel.tsx (50 lines modified)
• src/lib/soundEffects.ts (52 lines removed)

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5. Estate Menagerie Showcase & Prehistoric Habitat Modal [*763 lines*]

Overview: A dedicated sanctuary menagerie showcase displaying prehistoric creatures, habitat comfort ratings, paleo-temperature requirements, and global aura bonuses (e.g. Dimetrodon Geothermal Radiance, Mammoth Permafrost).
Reason for Removal: All creature inspection, genetics, breeding, and habitat controls were consolidated into PrehistoricSanctuaryPanel.tsx, rendering this separate modal duplicate code.

Archived Files:
• src/components/game/EstateMenagerieModal.tsx (763 lines)

6. Emergency Crisis Defense & Countdown Timer Suite [*727 lines*]

Overview: A standalone crisis triage and defense modal suite designed for sudden disasters (locust swarms, flash floods, plasma storms). Featured sandbag defenses, emergency triage actions, audio sirens, and an animated countdown clock.
Reason for Removal: The event system was migrated to the non-intrusive Event Engine notification system and dynamic weather widget.

Archived Files:
• src/components/events/CrisisDefenseModal.tsx (271 lines)
• src/components/events/EventEmergencyModal.tsx (249 lines)
• src/components/events/EventTickingClock.tsx (119 lines)
• src/components/events/EventEmergencyBanner.tsx (88 lines)

7. High-Stakes Agricultural Commodity Contract Deck [*578 lines*]

Overview: A card-deck UI offering timed, high-reward futures contracts for bulk crop deliveries with severe forfeit penalties, fluctuating market multipliers, and merchant guild standing.
Reason for Removal: Unused by the main interface; commercial contracts were merged into standard barn orders and artisan trade.

Archived Files:
• src/components/game/HighStakesContractDeck.tsx (578 lines)

8. Developer WebSocket Telemetry & Flood Surge Test Harnesses [*568 lines*]

Overview: Experimental WebSocket server/client prototype for real-time telemetry streaming and an offline hydraulic surge stress-test script.
Reason for Removal: These were standalone development artifacts located outside the core Next.js production build.

Archived Files:
• scripts/test_overflow_surge.ts (234 lines)
• examples/websocket/frontend.tsx (196 lines)
• examples/websocket/server.ts (138 lines)

9. Analog Hydrology Telemetry & Offline Return Catchup [*448 lines*]

Overview: Circular needle gauge dials displaying canal pressure and head levels, alongside an offline return dialog simulating water accumulation during inactivity.
Reason for Removal: Modern hydrology telemetry was merged directly into the top bar header and canal inspection cards.

Archived Files:
• src/components/game/EstateHydrologyGauge.tsx (251 lines)
• src/components/game/OfflineHydrologyReturnModal.tsx (197 lines)

■ How to Access or Restore Any Feature

All 22 archived files and their complete historical implementations are preserved with full commit history.

- **To view the branch on GitHub:** https://github.com/abhi543/meadowlands/tree/archive/dead-code-backup
- **To checkout the archive locally:** git checkout archive/dead-code-backup
- **To cherry-pick a specific file back into main:**
  git checkout archive/dead-code-backup -- src/components/game/EstateDispatchCenterModal.tsx